Tu (John) H. Nguyen

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EDUCATION

Drexel University, Philadelphia PA — Graduating June 2027
Bachelor of Science in Computer Science; Minor in Data Science and Financial Technology — **GPA: 3.8 - Dean's List**

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, HTML, CSS, C, C++, Rust, Bash
Libraries: NumPy, Pandas, Matplotlib, SQLAlchemy, jQuery, Bootstrap, Thymeleaf, Streamlit, FastAPI, ReactJS, Tokio
Frameworks/ Runtimes: Spring Boot, LangChain, LangGraph, Node.js, TailwindCSS, JUnit, OAuth 2.0
Databases: Microsoft SQL Server, PostgreSQL, PGVector, SQLite, Redis
Software Development: Agile Methodologies, Git, Gitflow, Large Language Models, System Design, RESTful API, Docker

RESEARCH EXPERIENCE

Drexel University, School of Computer and Information Sciences– Philadelphia, PA — October 2025–Present
Undergraduate Research Assistant - Software Engineering and Analytics Research (SOAR) Lab
- Owned end-to-end development of **TraceView**, an interactive visualization tool for agentic program repair trajectories; developed **Python**, **Jupyter Notebook**, and **Streamlit** tooling for trajectory annotation, graph-based visualization, filtering, metrics, and failure analysis, and contributed to the paper's methodology section by documenting the system design and implementation.
- Contributed to **AXE**, a multi-agent exploit engine for confirming zero-day vulnerability reports, supporting agentic vulnerability validation and exploitability analysis; co-authored the **AXE** paper under review and am contributing to a fourth manuscript in preparation

Drexel University, School of Computer and Information Sciences – Philadelphia, PA — July 2024–October 2025
Undergraduate Research Assistant
- Developed a simulation framework for InterPlanetary Archival Record Objects (**IPAROs**) in **IPFS/IPNS**, using **Bash** automation and Python (**NumPy**, **Pandas**) to evaluate 11 decentralized linking strategies through large-scale benchmarking
- Built data collection and visualization pipelines using **Matplotlib** to analyze storage efficiency, retrieval time, and scalability of IPARO implementations across thousands of simulated versions, validating theoretical complexity bounds with empirical results; presented findings at **ACM/IEEE Joint Conference on Digital Libraries 2025**

SELECTED PUBLICATIONS

- Patrick Le, Quang Bui, Alexander Grigorian, Alexey Kuraev, Thiyazan Salman, **Tu H. Nguyen**, Mat Kelly, Sawood Alam. "Linking Policies for the Permanent Web." ACM/IEEE Joint Conference on Digital Libraries 2025. DOI: 10.1109/JCDL67857.2025.00033
- Amirali Sajadi, **Tu H. Nguyen**, Kimmie Huynh, Esteban Parra, Preetha Chatterjee. "TraceView: Interactive Visualization of Agentic Program Repair Trajectories." IEEE International Conference on Software Maintenance and Evolution 2026 (presented).
- "AXE: Grey-Box Exploitability Confirmation for Localized Vulnerability Reports." Under review.

PROJECTS

Super Streaming: Low-latency Pub/Sub System — June–August 2024
- Designed and developed a high-performance pub/sub system in **Rust** using **Tokio**, achieving **sub-millisecond end-to-end latency** with up to 1k clients and sub-30ms latency with up to 10k clients, while ensuring data throughput of up to **500,000 messages/second**
- Created a multithreaded, MQTT v5-inspired system, purpose-built for streaming stock data from two exchanges, optimized for performance and scalability

iAgent : AI-Driven Project Assistance Tool — April–June 2024
- Designed and developed a full-stack internal application using **FastAPI**, **SQLAlchemy**, **PostgreSQL**, **React**, and **TailwindCSS** to support AI-assisted project planning and document workflows
- Built core platform features including a responsive multi-page frontend, REST API, **PGVector**-based vector store, and user authentication via **JWT** and **OAuth 2.0**
- Implemented a document management and **RAG** workflow using **OpenAI embeddings**, enabling users to organize workspace files and generate project requirements, test cases, and market research outputs

WORK EXPERIENCE

Drexel University, School of Computer and Information Sciences – Philadelphia, PA — Sep 2026–Present
Course Assistant - CS 281: Systems Architecture; SE 201: Intro to Software Engineering and Development
- Hold office hours and assist students during lab sessions by answering questions and guiding them through coursework
- Grade assignments and provide feedback to support student learning

Drexel University Information Technology, Core Enterprise Systems – Philadelphia, PA — Apr–Sep 2026; Apr–Sep 2025
Junior Developer
- Engineered new features and microservices for the **DrexelOne** portal, a production system serving thousands of students, faculty, and staff; implemented scalable integrations with **Microsoft SQL Server** using **Java**, **Spring Boot**, and **REST APIs**
- Built responsive web interfaces and faculty-facing applications using **jQuery**, **Bootstrap**, and **Thymeleaf**, translating specifications into production features used across university systems

SSI Securities Corporation – Ho Chi Minh City, Vietnam — April–September 2024
Software Engineer Intern
- Designed and developed various internal tools leveraging **Generative AI** to enhance customer care, streamline operations, and accelerate project development
- Engaged with cross-functional teams to diagnose and rectify critical issues in active projects using the **Gitflow** branching model, leading to streamlined workflows and enhanced project efficiency